Surface-Curvature-Distribution Effects on Turbine-Cascade Performance

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The manufacturing accuracy of turbine blades is judged by the difference between the designed and produced location of each point around the airfoils. This paper shows that the design-point and off-design-point aerodynamic and heat transfer performance of turbine cascades is affected significantly by the distribution of curvature along the suction and pressure surfaces, and less significantly by the exact location of each one point. The performances of turbine blades designed with continuous slope of curvature distribution, and with slight discontinuities in the slope of curvature, are examined using a computer program for various design and off-design flow conditions. The accuracy of the computations is checked by comparison of computational results with cascade solutions of known experimental output. It is concluded that the accuracy in local curvature is just as important as the design-location-by-manufactured-location accuracy of each point on the blade.

Introduction

Gas turbine blades are three-dimensional objects operating in a complex and unsteady flow field. Three-dimensional turbine blades are generated by "stacking" two-dimensional designs using rules for: the flow angles from hub to tip at the leading and trailing edge, derived from radial-equilibrium throughflow calculations; the locus of the centers of gravity of the sections; and the three-dimensional shape of the leading edge and the trailing edge. Compromises in performance must be made to accommodate these three-dimensional constraints, the location of cooling passages, and hollow sections. Due to its complexity the blade-design problem is usually reduced to a series of "stacked" two-dimensional problems, assuming that the cascade operates in a steady two-dimensional flow field.

Blade design is partly a science and partly an art. The performance is determined by the behavior of the boundary layers along the surfaces, and is expressed in the form of some property distribution (such as pressure, velocity, etc.) on the blade surface and in the cascade passage. Blade cascades can be designed by direct or inverse methods. In the direct method the designer inputs the geometry of the blade and the output is the performance of the airfoil. The blade geometry is modified until a desirable performance is obtained. In the inverse method the designer usually starts from some initial blade shape and performance, and inputs the desired modifications to the performance. The output is a new geometric shape and its performance, which is as close to the desired (input) one as permitted by the equations modeling the flow. Both methods are iterative and are based on the assumption of steady flow conditions.

The accuracy of the produced blade shapes is determined by comparing the intended (designed) shape with manufactured blades. The usual criterion is the allowable tolerance between design and manufactured points. The purpose of this paper is to show that comparisons of design-versus-manufactured curvature distribution at each point should be used in addition to point-to-point comparisons as a criterion for accepting an airfoil. Two sets of sample cascades are used. The cascades of the first set have similar shapes, with small local slope-of-curvature discontinuities. The latter result in small surface variations that would pass the point-for-point-tolerance tests, but introduce unacceptable cascade performance. The cascades of the second set have substantial surface variations among them, but they have been designed with continuous-slope-of-curvature surfaces. The second set is used to show that curvature distribution is what governs the shape of the surface Mach number and pressure distributions, which in turn determine boundary-layer behavior and blade performance.

Importance of Surface Curvature

Both the curvature and the slope of curvature of blade surfaces affect boundary-layer development and blade performance. The flow in turbomachinery cascades *curves* around the blades. The compressible-flow Navier–Stokes equations for Newtonian fluids in the limiting case of polar coordinates (r, θ) (taken from cylindrical-coordinate equations in the text by Bird et al., 1960) show the dependence of the core flow and the stress terms τ_{ij} on the local radius of curvature r:

$$\frac{\partial \rho}{\partial t}+\frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r)+\frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta)=0$$

Contributed by the International Gas Turbine Institute and presented at the 37th International Gas Turbine and Aeroengine Congress and Exposition, Cologne, Germany, June 1–4, 1992. Manuscript received by the International Gas Turbine Institute February 19, 1992. Paper No. 92-GT-84. Associate Technical Editor: L. S. Langston.

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$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}-\frac{v_\theta^2}{r}\right)=$$
$$-\frac{\partial p}{\partial r}-\left(\frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rr})+\frac{1}{r}\frac{\partial \tau_{r\theta}}{\partial \theta}-\frac{\tau_{\theta\theta}}{r}\right)+\rho g_r$$
$$\rho\left(\frac{\partial v_\theta}{\partial t}+v_r\frac{\partial v_\theta}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r v_\theta}{r}\right)=$$
$$-\frac{1}{r}\frac{\partial p}{\partial \theta}-\left(\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\tau_{r\theta})+\frac{1}{r}\frac{\partial \tau_{\theta\theta}}{\partial \theta}\right)+\rho g_\theta \quad (1)$$
$$\tau_{rr}=-\mu\left[2\frac{\partial v_r}{\partial r}-\frac{2}{3}(\nabla\cdot\mathbf{v})\right]$$
$$\tau_{\theta\theta}=-\mu\left[2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r}{r}\right)-\frac{2}{3}(\nabla\cdot\mathbf{v})\right]$$
$$\tau_{r\theta}=\tau_{\theta r}=-\mu\left[r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta}\right]$$
$$(\nabla\cdot\mathbf{v})=\frac{1}{r}\frac{\partial}{\partial r}(rv_r)+\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}$$

The stress components τ_{ij} are dominant in large-shear areas, such as the boundary layer in steady two-dimensional cascade flows. The $1/r$ and $1/r^2$ terms in these equations suggest a strong dependence of local pressure and velocity distributions on curvature. While one expects that the boundary layer shields the core of the flow from surface discontinuities, the boundary-layer behavior depends on the slope of the velocity distribution as it approaches the solid wall, and Eqs. (1) indicate that the slope of the velocity distribution, the boundary layer, and its behavior, are coupled with the local radius. Smooth velocity distributions along the blade surfaces require smooth surface-curvature distributions (continuous slopes of velocity and curvature along the blade surface). Continuous slope of curvature requires continuous third derivatives, as illustrated in the following two equations for curvature and slope of curvature for $y = f(x), y' = df(x)/dx, y'' = d^2f(x)/dx^2$ and $y''' = d^3f(x)/dx^3$.

$$\mathbf{C}(x)=\frac{1}{R(x)}=\frac{y''}{[1+y'^2]^{(3/2)}} \quad (2)$$

$$\mathbf{C}'(x)=\frac{d\mathbf{C}}{dx}=\frac{y'''[1+y'^2]-3y'y''^2}{[1+y'^2]^{(5/2)}} \quad (3)$$

Most parametric splines currently in use for direct blade design (e.g., cubic, B-splines, Bezier splines, etc.) have continuous first and second derivatives, and they result in smooth-looking surfaces with continuous curvatures, but discontinuous slopes of curvature at the spline knots. One must distinguish here between the small local slope-of-curvature discontinuities from surface roughness and fouling, with which turbines must operate, and the slope-of-curvature discontinuities in the as-designed shape at the junctions of the splines. It will be shown below that the latter are invisible to the eye, but they may produce unusually loaded cascades and thicker wakes. This has resulted in many test and production airfoils (isolated and in turbomachine cascades) that exhibit Mach number and pressure-coefficient-distribution spikes or dips and humps of various magnitudes. These occasionally result in unexpected loading distributions along the blade length, and in local separation bubbles. These effects are visible as small local "kinks" in surface pressure or Mach number distributions in some of the computational and experimental data published, for example, by: Okapuu (1974), Gostelow (1976), Wagner et al. (1983), Hourmouziadis et al. (1987) (with separation bubble), Sharma et al. (1992) and others.

Blending a leading-edge circle or ellipse with the blade surfaces usually results in a local curvature or slope-of-curvature discontinuity. This may cause separation and substantial leading-edge surface-Mach-number spikes, which are detected only if the probes are located at the correct location. Such a local leading-edge laminar-separation bubble due to blending of a leading-edge circle with the blade surfaces occurs in the turbine geometry published by Hodson and Dominy (1987), seen in the data published in Fig. 11 of Hodson (1985). This leading-edge spike and separation region were removed by modifying the geometry of the blade in the vicinity of the slope-of-curvature discontinuity with an inverse design technique as explained in Figs. 11, 12, and 13 of Stow (1989).

A direct-design method (Korakianitis, 1989) using analytic polynomials indicated that continuous-slope-of-curvature distribution was essential to avoid local surface-pressure-distribution spikes and "kinks." A subsequent direct-design method using parametric splines specified with continuous slope of curvature (Korakianitis and Pantazopoulos, 1993a) eliminated unwanted performance "kinks" but proved cumbersome and time consuming. A third direct-design method (Korakianitis, 1993b) controlled the local and global acceleration (and diffusion) by prescribing the curvature distribution along the blade length ($\mathbf{C}(x)$) such that the slope of curvature is continuous along the blade surfaces. The blade geometry was computed by mapping the input curvature in the (x, y) plane. In the last method the trailing-edge region is specified by an analytic polynomial, the main part of the blade surface is specified by its curvature distribution, and the leading-edge region is specified by a thickness distribution evaluated with respect to a construction line, such that it joins the blade surface with continuous slope of curvature. Using this method we showed

Nomenclature

b = axial chord length (nondimensionally $b = 1$)
c = blade chord
$\mathbf{C}$ = $1/R$ = curvature (Eq. (2), as shown in Fig. 3*b*)
C_L = tangential-loading coefficient = $2S\cos^2\alpha_2(\tan\alpha_1 - \tan\alpha_2)/b$
g = body force (Eqs. (1))
i = flow incidence with respect to α_1
M = Mach number
o = throat diameter of the cascade
p = pressure (Eqs. (1))
R = local radius of curvature (Eq. (2))
s = blade-surface length
S = tangential spacing (pitch) of the blades
t = time (Eqs. (1))
v_i = velocity component ($i = r, \theta$)
$\mathbf{v}$ = velocity vector (Eq. (1))
(x, y) = Cartesian coordinates
(X, Y) = nondimensional distances $(X = x/b, Y = y/b)$
α = flow angle
δ_1 = boundary-layer displacement thickness (Figs. 3*c* and 5)
λ = stagger angle of the blades
μ = dynamic viscosity (Eqs. (1))
ρ = density (Eqs. (1))
τ_{ij} = shear stress components (Eqs. (1), $i, j = r, \theta$)

Subscripts

1, 2 = cascade inlet and outlet
r = radius in polar coordinates (r, θ)
θ = angle in polar coordinates (r, θ)

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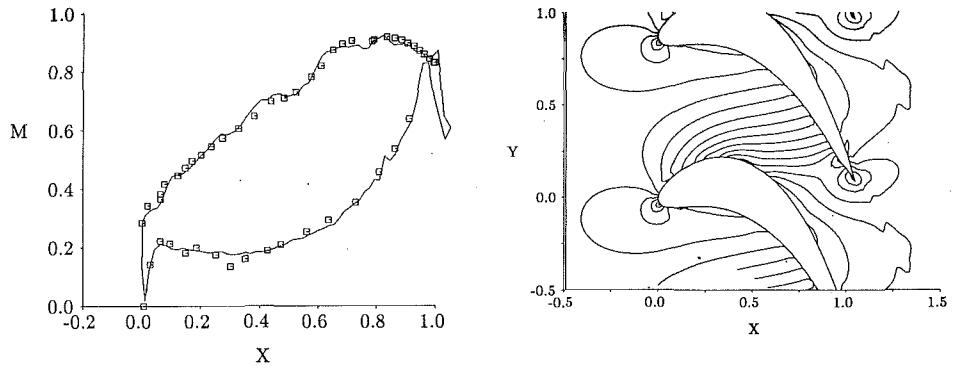


Fig. 1 Comparison of Mach number distributions for the turbine cascade investigated by Kiock et al. (1986). Left, surface Mach-number distributions. Right, cascade geometry and Mach number contours of increment 0.05.

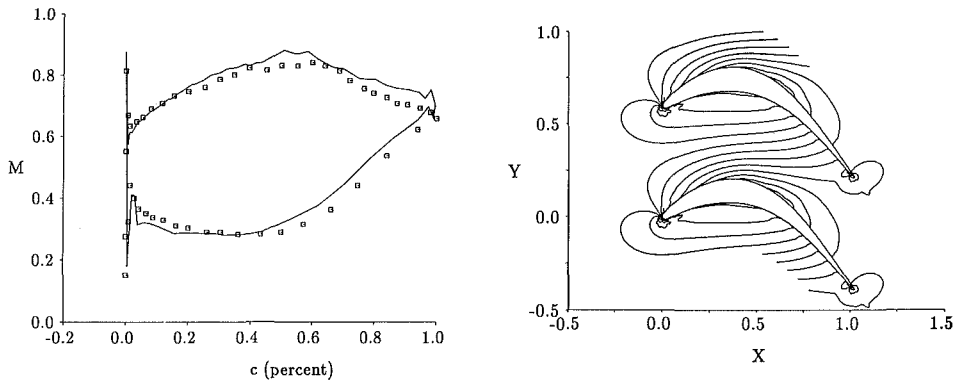


Fig. 2 Comparison of Mach number distributions for the turbine cascade investigated by Hodson and Dominy (1987). Left, surface Mach-number distributions. Right, cascade geometry and Mach number contours of increment 0.05.

that increasing the stagger angle produces more-front-loaded cascades, and that intermediate values of stagger angle promote intermediate-loading philosophy and result in thinner wakes.

Accuracy of Computations

The performance of sample turbine blades in this paper has been evaluated using the inviscid and steady-flow version of Giles' (1988) computer program UNSFLO. The accuracy of the computations has been checked by comparing the results of Euler computations with two cases of known theoretical output (Korakianitis, 1993b). The first case was a comparison of computations with the (exact) theoretical solution of Gostelow's (1984) compressor cascade. The second case was a comparison of computations with the hodograph solution of Hobson's transonic turbine cascade No. 2 (taken from AGARD, 1985). These tests were suitable for two-dimensional inviscid flow fields with sharp trailing edges, and indicated exact agreement between theory and computation. In this paper we show two comparisons with experiments.

The first case is a comparison of inviscid computations with the experimental results from the subsonic turbine cascade obtained at DFVLR Braunschweig by Kiock et al. (1986). For this case $M_1 = 0.260$, $M_2 = 0.782$, $\alpha_1 = 30.0$ deg and $\alpha_2 = -67.33$ deg. A uniform **H**-grid of 100 × 40 and a cusp (to close the trailing edge) were used. The results, shown in Fig. 1, indicate very good agreement with the experiments. The tick marks are the experimental results (transformed from data along percentage of chord c to data along percentage of axial chord b, along x), and the solid line is the calculation. Despite the absence of boundary-layer computations or corrections for it, the numerical solution follows local dips in the suction-surface Mach number distribution at $x \approx 0.55$ and $x \approx 0.80$, and local accelerations at $x \approx 0.05$ and $x \approx 0.70$.

The second case is a comparison of inviscid computations with the experimental results from the subsonic turbine cascade obtained at Cambridge University by Hodson and Dominy (1987). For this case $M_1 = 0.496$, $M_2 = 0.710$, $\alpha_1 = 38.8$ deg and $\alpha_2 = -53.5$ deg. A uniform **H**-grid of 150 × 60 and a cusp (to close the trailing edge) were used. The results, shown in Fig. 2, indicate very good agreement of computations with the experimental data (tick marks) taken from Stow (1989). Hodson (1985) has analyzed the experimental data to show leading-edge laminar-separation bubbles on both surfaces, caused by blending a leading-edge circle with the blade surfaces. The numerical solution predicts the leading-edge surface Mach number spikes. Despite the three-dimensional effects in the experiments, the effect of the trailing edge cusp in the trailing-edge region of the solution, and the absence of boundary-layer computations or corrections for it, the agreement between computation and experiment is very good.

The First Set of Sample Cascades

Figure 3 shows the geometry, surface-curvature distribution, and surface Mach number distribution of four very similar turbine cascades designed for $\alpha_1 = 40$ deg, $\alpha_2 = -60$ deg, $M_2 = 0.8$, $C_L = 1.0$, and $\lambda = -32$ deg. C_L is the tangential-loading coefficient, defined as tangential blade loading, divided by tangential blade area, divided by outlet dynamic head. For incompressible flow C_L is reduced to the function of S/b, α_1, and α_2 (Wilson, 1984) shown in the nomenclature. For these cascades $S/b = 0.7779$. These four blades have been designed using the method described by Korakianitis (1989). The pressure surfaces are identical. The suction surfaces have minute differences as described below. In that method the input was the (X, Y) points of the knots of surface segments and surface angles near the leading and trailing edges. The surface

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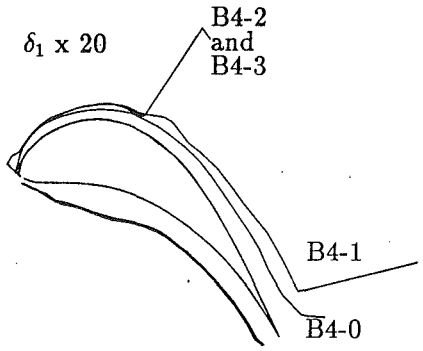


Fig. 3(*a*) Blade geometries and boundary layer displacement thickness δ_1 (free-stream turbulence 15 percent)

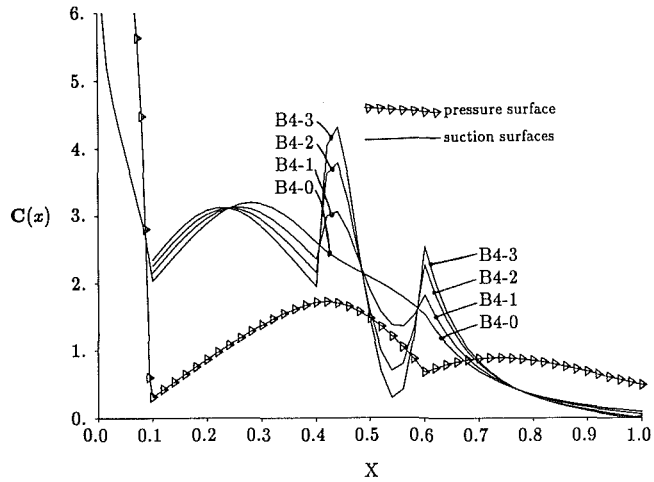


Fig. 3(*b*) Curvature distributions

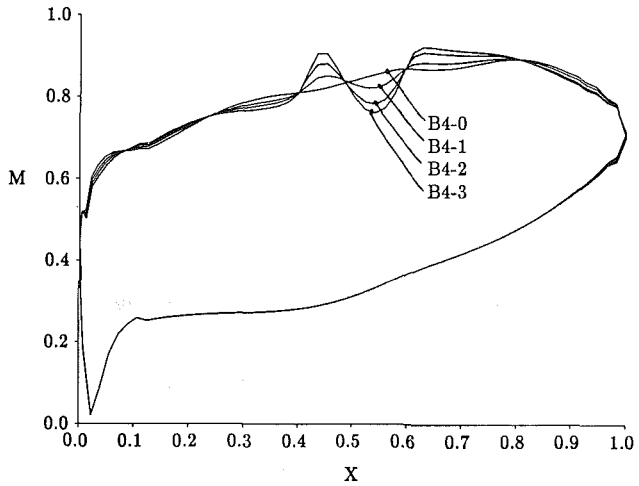


Fig. 3(*c*) Surface Mach number distributions

Fig. 3 Surface distributions for cascades B4-0, B4-1, B4-2, B4-3: α_1 = 40 deg, α_2 = −60 deg, C_L = 1.00, λ = −32 deg

segments had point, first-derivative, and second-derivative continuity at the knots. (These conditions result in continuous curvature distribution along each blade surface, but in general in slope-of-curvature discontinuity at the segment knots.) For the sample cascades shown, two consecutive knots are at X = 0.40 and X = 0.60. As shown in Fig. 3(*b*), cascade B4-0 has a smooth suction-surface curvature $\mathbf{C}(x)$ (without slope-of-curvature discontinuity at the knots at X = 0.40 and X = 0.60). Cascades B4-1, B4-2, and B4-3 have continuous suction-surface curvature ($\mathbf{C}(x)$) distributions, but they have progressively worse slope-of-curvature ($\mathbf{C}'(x)$) discontinuities at X = 0.40 and X = 0.60. These suction-surface curvature distributions result in the surface Mach-number distributions shown in Fig. 3(*c*). The smoothness in Mach-number distribution for cascade B4-0, and the spikes and dips in Mach-number distribution for cascades B4-1, B4-2, and B4-3 are "mapped" from the corresponding surface-curvature distributions.

Quantitatively the nondimensional differences in the y locations of corresponding blade points are four orders of magnitude smaller than the axial chord. Therefore, the suction-surface differences between the four blades are not visible in Fig. 3(*a*), where the four blades are plotted on top of each other. The premise of this paper is that surface-curvature distribution *along the entire blade length*, as well as point-for-point surface accuracy, are important for Mach number and pressure distributions, and the resulting boundary-layer behavior and cascade performance. Thus the differences in curvature distributions along the length of the four blades shown in Fig. 3(*b*) (which start at X = 0.1 and end at X = 1.0, but they are dominant in the sample cascades at $0.4 < X < 0.6$), result in surface-curvature and Mach number dips and spikes as shown in Fig. 3(*c*). Figure 3(*a*) shows the corresponding boundary-layer displacement-thickness distribution $\delta_1 \times 20$ of each blade, calculated with Albers and Gregg's (1974) differential boundary layer program. These calculations were performed assuming incoming free-stream turbulence of 15 percent and Reynolds number 5.5×10^5 based on chord. This program recognizes the importance of surface-curvature distribution, which is part of the input for the computations. Blade B4-0 has smooth curvature and Mach number, and results in the thinnest boundary layer. Blade B4-1 has a small discontinuity in the slope of curvature that causes a disturbance in surface Mach number distribution, and a thicker boundary layer (and wake) at the trailing edge. Blades B4-2 and B4-3 have correspondingly larger slope-of-curvature discontinuities, that cause the boundary layer to separate (similarly to the leading-edge separation shown in Fig. 2) near the discontinuity.

Boundary Layer Calculations

For Incidence i = 0 deg. Figures 4(*a*) and 4(*b*) show the Mach number distributions for cascades B4-0 and B4-1 respectively for incidences of +10, 0, and −10 deg with respect to the design inlet-flow angle α_1. Figure 4(*c*) shows the corresponding boundary-layer displacement thickness δ_1 calculated with Albers and Gregg's (1974) program, with *specifications identical with those for Fig. 3(a). For zero incidence* (i = 0 deg) the suction-surface boundary-layer displacement thickness of cascade B4-0 is smaller than that of cascade B4-1, as already expected from Fig. 3(*a*).

For Positive Incidence i = +10 deg. This increases the suction-surface Mach number and decreases the pressure-surface Mach number at $X \approx 0.05$. This front loads both cascades substantially, and at i = +10 deg the program indicates that both cascades B4-0 and B4-1 experience suction-surface laminar separation in the region of sudden deceleration at $X \approx 0.05$. This is followed by immediate turbulent reattachment, which in turn is followed by immediate turbulent separation, resulting in thick wakes at the trailing edge.

For Negative Incidence i = −10 deg. This decreases the suction-surface Mach number and increases the pressure-surface Mach number at $X \approx 0.05$. This results in substantially different behavior between cascades B4-0 and B4-1.

For blade B4-0: For i = 0 deg the boundary layer of B4-0 exhibited normal transition between $X \approx 0.65$ (onset of transition) and $X \approx 0.93$ (fully turbulent) without separation.

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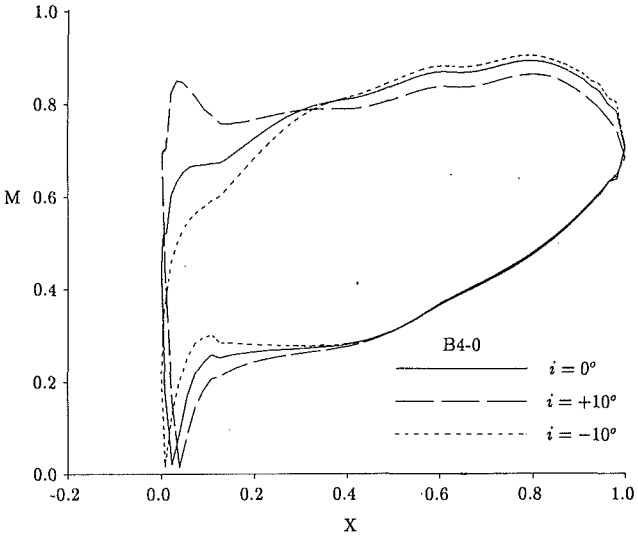


Fig. 4(*a*) Surface Mach number distributions at ±10 deg incidence for cascade B4-0

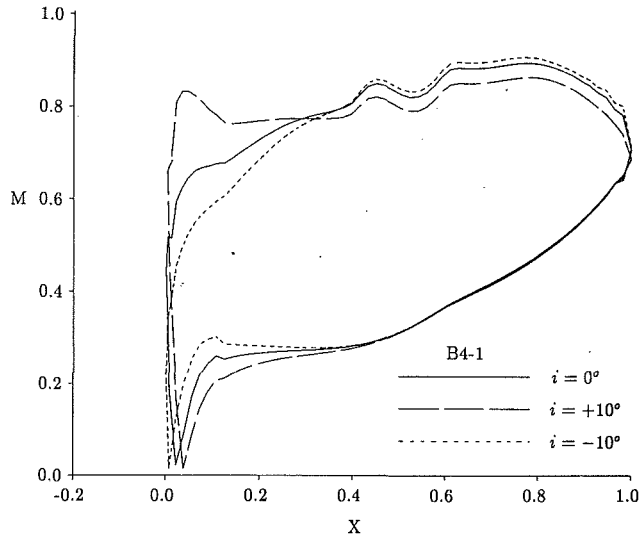
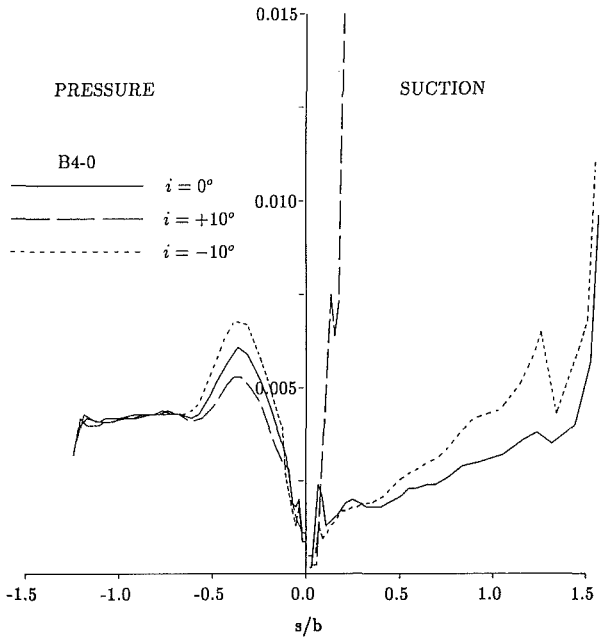


Fig. 4(*b*) Surface Mach number distributions at ±10 deg incidence for cascade B4-1



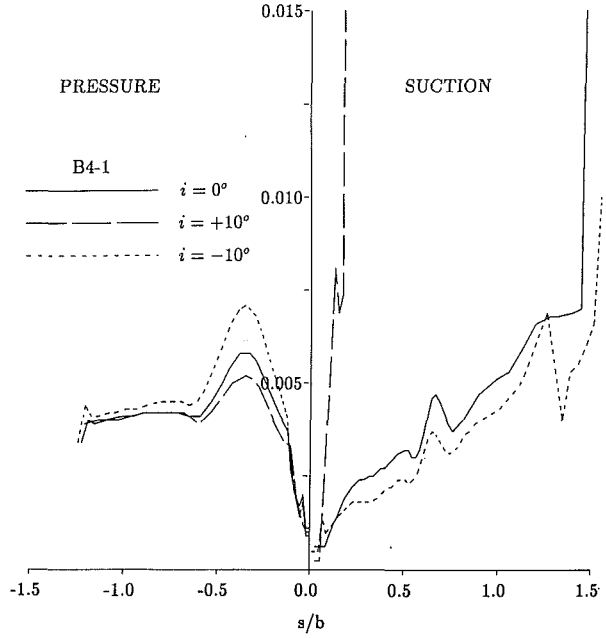


Fig. 4(*c*) Boundary layer displacement thickness δ_1 for cascades B4-0 and B4-1 at ±10 deg incidence (free-stream turbulence 15 percent)

Fig. 4 Surface Mach number distributions and boundary layer displacement thickness δ_1 for cascades B4-0 and B4-1 at ±10 deg incidence

For $i = -10$ deg the onset of transition appeared a little further downstream (at $X \approx 0.70$). At that point the boundary layer was relatively thicker compared to that at the onset of transition for $i = 0$ deg. The small delay in transition is because of the smoother and less rapid velocity acceleration near the leading edge. The thicker boundary layer could not withstand the subsequent adverse pressure gradient, and for $i = -10$ deg the code showed transitional separation at $X = 0.87$. With a turbulent restart option the program did not show any further separation, but resulted in a thicker wake compared to that for $i = 0$ deg.

For blade B4-1: For $i = 0$ deg the boundary layer of B4-1 exhibited normal transition between $X \approx 0.773$ and $X \approx 0.93$, associated with turbulent separation at $X \approx 0.98$. For $i = -10$ deg (unlike the boundary-layer behavior of blade B4-0), the considerable disturbances in Mach number distribution in the region $0.4 < X < 0.6$ caused blade B4-1 to experience earlier onset of transition, at $X \approx 0.68$. At that point the boundary layer was substantially thinner compared to that at the onset of transition for $i = 0$ deg. As a result the program indicates fully developed turbulent flow at $X \approx 0.85$ (a little earlier than in the corresponding case for blade B4-0), resulting in a thinner wake than for $i = 0$ deg for blade B4-1.

The Second Set of Sample Cascades

Figure 5 shows five continuous-slope-of-curvature cascades designed for $\alpha_1 = 40$ deg, $\alpha_2 = -60$ deg, $M_2 = 0.8$, $C_L = 1.0$, $\lambda = -25$ deg, and $o/S = 0.5$ ($\approx \cos\alpha_2$) using the prescribed-curvature-distribution direct-design method described by Korakianitis (1993b). The throat diameter is an input in this design method, and $S/b = 0.7779$. The suction surfaces of all five cascades pass from a common region at the throat. All suction surfaces must pass through the same origin at the

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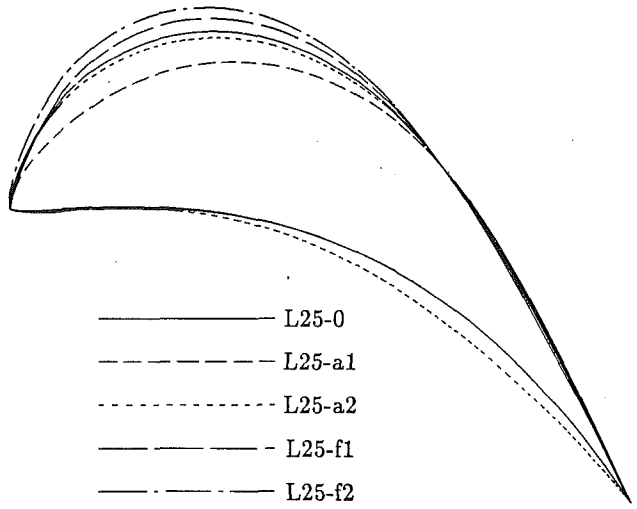


Fig. 5(*a*) Geometries of cascades L25-0, L25-f1, L25-f2, L25-a1, and L25-a2

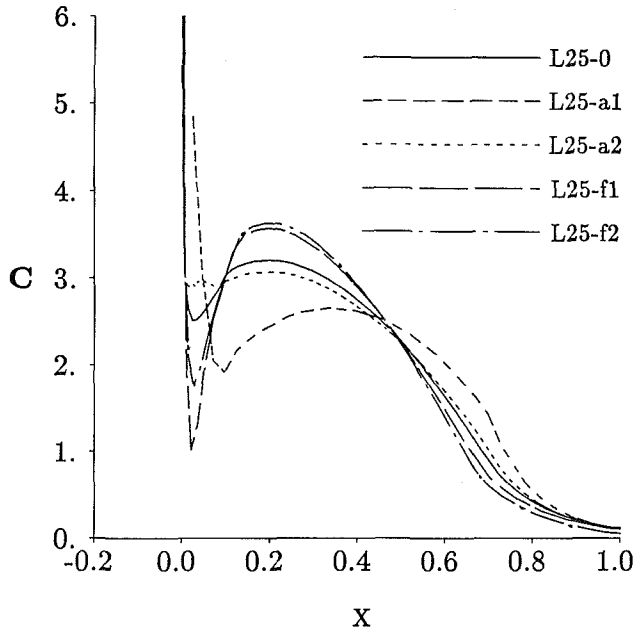


Fig. 5(*b*) Surface curvature (C(*x*)) distributions

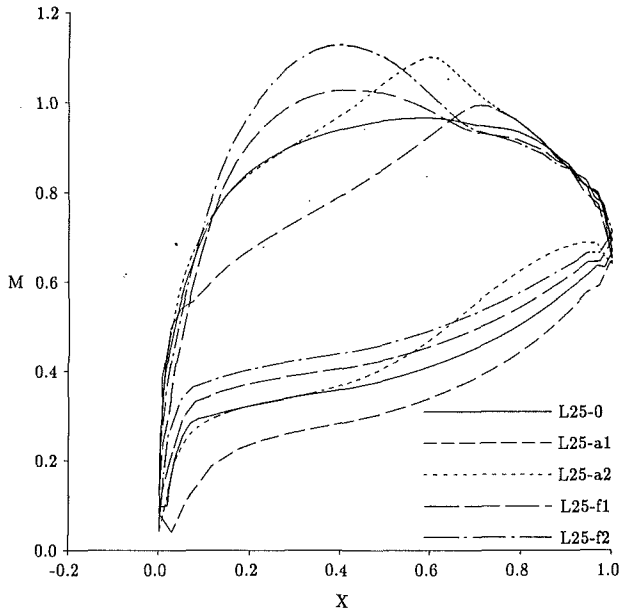


Fig. 5(*c*) Surface Mach number distributions

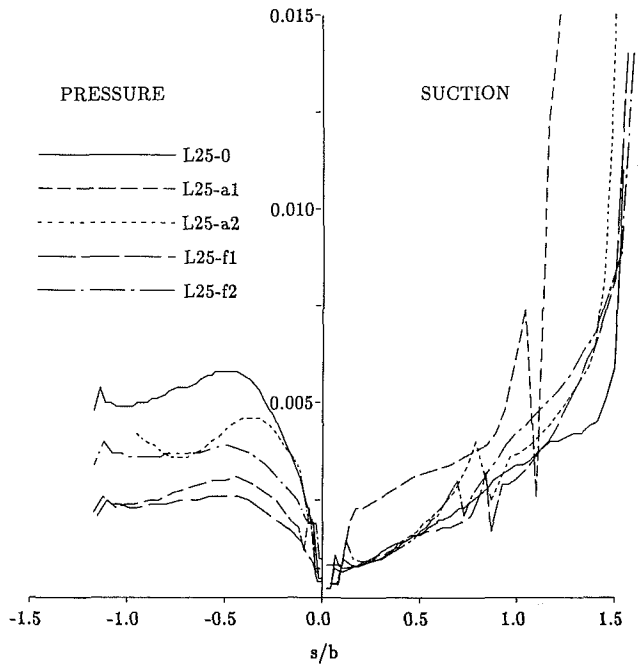


Fig. 5(*d*) Boundary layer displacement thickness δ_1

Fig. 5 Surface distributions for cascades L25-0, L25-f1, L25f2, L25-a1, L25-a2. $\alpha_1 = 40$ deg, $\alpha_2 = -60$ deg, $C_L = 1.00$, $\lambda = -25$ deg

leading edge. Those with lower curvature in the region of unguided diffusion (on the suction surface downstream of the throat) have higher suction surfaces cutting deeper into the passage. They also have higher curvatures in the corresponding suction-surface regions (in the passage) in order to reach the same origin, as indicated in Figs. 5(*a*) and 5(*b*). Thus by increasing (decreasing) the curvature distribution forward (versus aft) on the suction surface, the designer can front (versus aft) load the Mach number distribution. Cascade L25-0 has a smooth midloaded characteristic. Cascades L25-a1 and L25-a2 ("a" stands for aft loaded) are correspondingly more aft-loaded than L25-0. Cascades L25-f1 and L25-f2 ("f" stands for front loaded) are correspondingly more front-loaded than L25-0. The front versus aft loading has a direct correspondence with the suction surface curvature distribution, as indicated in Figs. 5(*b*) and 5(*c*).

Four of the cascades have identical pressure surfaces. Figure 5(*c*) shows that the pressure surface Mach numbers of the four identical pressure surfaces increase appropriately to match the suction-side loading. Comparing the pressure-surface blade shape and the pressure-side loading of cascade L25-a2 to those of the other four cascades indicates that lowering the pressure surface in the passage increases the pressure-surface Mach number (thus loading the suction-surface Mach number even more).

Figures 5(*a*), 5(*b*), and 5(*c*) illustrate that, as the curvature near the front part of the suction surface is increased, the cascade becomes front loaded; and as the curvature near the aft part of the suction surface is increased, the cascade becomes aft loaded. Curvature variations of the same magnitude have a correspondingly larger effect farther downstream on the surfaces, and they are critical in the region of unguided diffusion downstream of the throat on the suction surface. Higher values of stagger angle λ force the geometry such that the throat is located farther upstream on the suction surfaces. The trailing-

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edge regions of the blades are located in lower Y values in the (X, Y) plane. Thus the blades become thinner, with less curvature near the trailing edge, and correspondingly more curvature near the leading edge. The higher curvatures near the leading edges result in more front-loaded cascades for higher λ (Korakianitis, 1993b). The designer has more flexibility in controlling the front-versus-aft loading of a cascade with lower stagger angles.

The boundary-layer calculations illustrated in Fig. 5(d) have been performed with Albers and Gregg's (1974) program, with 15 percent free-stream turbulence, and Reynolds number 5.5 $\times 10^5$ based on chord. The aft-loaded cascades have relatively short surface lengths, with relatively more adverse pressure gradients toward the trailing edge. The front-loaded cascades have relatively long surface lengths, with relatively less adverse pressure gradients toward the trailing edge. This clearly affects the development of the pressure- and suction-surface boundary layers. The intermediate blade-loading philosophy (L25-0) results in the thinnest suction-surface wake. This agrees with the conclusions of a previous investigation by Hoheisel et al. (1987), which concluded that a midloaded cascade had a relatively thinner boundary layer, although their blades did not have the rest of the design parameters (such as throat diameter) identical. The front-loaded cascades L25-f1 and L25-f2 have thicker suction-surface wakes than that of cascade L25-0, but thinner than those of the aft-loaded cascades L25-a1 and L25-a2. The two aft-loaded cascades have lower suction-surface local Reynolds numbers, which result in larger laminar boundary-layer thicknesses, especially for cascade L25-a1. This cascade exhibits transition immediately after the peak in Mach number distribution at $X \approx 0.78$ ($s/b \approx 1.1$). The boundary layer cannot withstand the severe adverse pressure gradient and it eventually separates at $s/b \approx 1.3$. Another reason for this behavior is the relatively sudden acceleration of the flow in the leading-edge region. Although cascade L25-a2 is aft-loaded even more than cascade L25-a1, the relatively higher suction-surface Reynolds numbers, and the less sudden acceleration of the flow in the leading-edge region, result in thinner laminar boundary layers. The onset of transition is earlier, at $X \approx 0.57$ ($s/b \approx 0.85$), and the flow does not separate until the trailing edge.

Comparison of the Mach number distributions shown in Figs. 2, 3, 4, and 5 indicate that continuous-slope-of-curvature cascades can have substantial variations in geometric blade shapes that result in acceptable performance. Small slope-of-curvature discontinuities result in small surface variations, that may pass the manufacturing-tolerance tests, yet introduce unacceptable performance (and in some cases regions of undesirable separation).

Other Comments

In the early stages of axial-cascade design the geometry of airfoils is analyzed in a series of two-dimensional cuts from hub to tip. A major effort in the subsequent three-dimensional design process is to minimize the radial pressure gradients, radial flows, and other secondary flows. These three-dimensional effects always exist in actual flows, and they are particularly important in low hub-to-tip-ratio stages. Although three-dimensional design and analysis programs are widely used, much of our basic understanding of axial-turbine flows (derived from theory, computation, and experiments) is based on two-dimensional considerations. Three-dimensional investigations are enlightening, useful, and more complete, but more expensive and harder to model than two-dimensional ones. Both two-dimensional and three-dimensional approaches are necessary for the analysis of physical phenomena in different phases of turbomachinery design. This paper investigates the effects of two-dimensional surface-curvature distribution on two-dimensional flows in axial turbines. It is expected that three-dimensional curvature has similar effects to the two-dimensional ones analyzed here. Of course theoretical and computational investigations, including the present one, would benefit from experimental verification.

Curvature and slope of curvature are nonlinear functions of point-by-point geometry (expressed as (X, Y) points, or as $y(x)$, see Eqs. (2) and (3)). Therefore small variations in geometry can cause large variations in curvature, and small variations in curvature can cause large variations in geometry; see Figs. 3 and 5. Performance is a function of curvature distribution along the entire airfoil length, not just the point-by-point variation in curvature. This should not be compared with the accuracy of numerical-control cutting machines or the tolerance criteria for point-by-point quality control of manufactured blades. New tolerance criteria for surface-curvature distribution (along the chordwise blade length) must be established.

Conclusions

Small slope-of-curvature discontinuities in the blade surfaces result in Mach number or pressure surface-distribution spikes or humps, that may introduce regions of local acceleration and deceleration, separation, or undesirable loading distributions along the length of the blades. Smooth-slope-of-curvature blade surfaces avoid local kinks and spikes in surface-pressure and surface-velocity distributions, and facilitate the design of lower-loss higher-loading cascades. Continuous-slope-of-curvature surfaces should be used wherever the aerodynamic, heat transfer, hydrodynamic, or aesthetic performance, appropriately defined in each case, is critical.

Blade surfaces with continuous slope of curvature throughout their length have more desirable performance and allow the designer considerable flexibility in controlling the front versus aft loading distribution of a cascade. It is found that the cascade performance, judged by the shape of the surface Mach number distribution, is extremely sensitive to small changes in slope-of-curvature distribution of the airfoil surfaces.

The performance of two sets of turbine cascades has been presented and their boundary-layer behavior has been numerically investigated. The first set includes four near-identical blade shapes with small slope-of-curvature discontinuities (resulting in blade-surface variations that would pass the point-tolerance criteria). The small local discontinuities in the slope of curvature introduce unacceptable performance for two of the four cascades. The second set of sample cascades includes five geometries and it shows that blades designed with continuous slope-of-curvature distributions, with substantial variations in blade geometries, result in cascades of various loading distributions that are acceptable. It is concluded that in addition to point-for-point tolerance, blade designs should be checked for their point-for-point design-versus-manufactured surface-curvature distributions.

Acknowledgments

The authors thank Professor Michael B. Giles and Rolls Royce plc. for their permission to use UNSFLO to compute the performance of the sample cascades and Mr. N. Vlachopoulos for the many helpful discussions during the course of the investigation.

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